

DEPARTMENT OF ELECTRICAL ENGINEERING
INSTRUMENTATION & MEASUREMENT — CLO 2 PROJECT REPORT

SINGLE-AXIS SOLAR TRACKING SYSTEM

LDR-BASED DIFFERENTIAL SENSING WITH ARDUINO CLOSED-LOOP CONTROL

MICROCONTROLLER	ACTUATOR	SENSORS	TRACKING AXIS
ARDUINO UNO R3	SG90 SERVO	2× LDR	SINGLE-AXIS

Group Members:
Maliha Semi Anwar EE-24004
Sania Tariq EE-24027
Fareena Azeem EE-24002
Maha Tariq EE-24023

1. OBJECTIVE

The objective of this project is to design and implement a single-axis solar tracking system that maximizes solar energy capture by continuously aligning a solar panel with the direction of maximum sunlight intensity.

This is achieved using Light Dependent Resistors (LDRs) as sensing transducers, a control unit (Arduino), and a servo motor for mechanical positioning. The system detects the variation in light intensity and automatically adjusts the panel angle to maintain optimal orientation.

The key goals include:

- To study the behavior of photoconductive transducers (LDRs)
- To implement a closed-loop control system using differential sensing
- To analyze the relationship between light intensity and resistance
- To improve solar panel efficiency compared to a fixed panel
- To validate the mathematical models governing sensor response and system control

2. INTRODUCTION

2.1 Overview of Solar Tracking System

Solar energy systems achieve maximum efficiency when solar panels are perpendicular to sunlight rays. However, due to the Earth's rotation, the sun's position changes continuously throughout the day.

A single-axis solar tracking system solves this problem by rotating the panel along one axis (usually east to west), ensuring that it always faces the sun at an optimal angle.

This project uses a sensor-based tracking method, where light intensity is measured and used as feedback to control panel movement.

2.2 Type of Transducer Used

The system uses Light Dependent Resistors (LDRs), which are passive photoconductive transducers.

Photoconductive Transducer Definition

A photoconductive transducer converts light energy into a change in electrical resistance.

LDR Characteristics

- Resistance decreases with increasing light intensity
- Very high resistance in dark ($M\Omega$ range)
- Low resistance in light ($k\Omega$ range)

This behavior is based on the principle of photoconductivity.

2.3 Working Principle of LDR

An LDR is made of semiconductor material (e.g., Cadmium Sulfide).

Principle: Photoconductive Effect

When light photons fall on the LDR:

- Energy is absorbed by electrons
- Electrons move from valence band to conduction band
- Number of free charge carriers increases
- Resistance decreases

Thus:

$$\text{Light Intensity } \uparrow \Rightarrow \text{Resistance } \downarrow$$

2.4 Mathematical Model of LDR

The relationship between resistance and illuminance is:

$$R(E) = R_0 \cdot E^{-\gamma}$$

Where:

- $R(E)$ = Resistance at illuminance E
- R_0 = constant
- γ = material-dependent constant
- E = Light intensity (lux)

Logarithmic Form (for graphing)

$$\log R = \log R_0 - \gamma \log E$$

This shows a **linear relationship on a log-log graph**, which validates experimental observations.

2.5 Voltage Divider Principle

Each LDR is connected in a **voltage divider circuit** to convert resistance into voltage.

$$V_{out} = V_{cc} \cdot \frac{R_{fixed}}{R_{LDR} + R_{fixed}}$$

Explanation

- When light increases $\rightarrow R_{LDR}$ decreases $\rightarrow V_{out}$ changes
- This voltage is read by Arduino analog pins

Thus, light intensity is converted into a measurable voltage signal.

2.6 Differential Sensing Mechanism

Two LDRs are placed on opposite sides of the panel.

- Left sensor $\rightarrow V_{A0}$
- Right sensor $\rightarrow V_{A1}$

The system calculates:

$$\Delta V = V_{A0} - V_{A1}$$

Interpretation

- $\Delta V = 0 \rightarrow$ Panel is perfectly aligned
- $\Delta V > 0 \rightarrow$ Rotate in one direction
- $\Delta V < 0 \rightarrow$ Rotate in opposite direction

This is called a **null-balance control system**.

2.7 Control System (Closed-Loop Operation)

The system operates as a **closed-loop feedback control system**:

1. Sensors detect light intensity
2. Arduino processes the difference
3. Error signal is generated (ΔV)
4. Servo motor adjusts panel angle
5. System continuously updates position

2.8 Servo Motor Control

The servo motor rotates the panel based on the error signal.

$$\theta = \text{map}(\Delta V, -5V, +5V, 0^\circ, 180^\circ)$$

Where:

- θ = Panel angle
- Mapping converts voltage to angular position

2.9 Power Output and Efficiency)

The solar panel output is:

$$P = V \times I$$

Tracking improves power output by maintaining optimal orientation.

Efficiency improvement is:

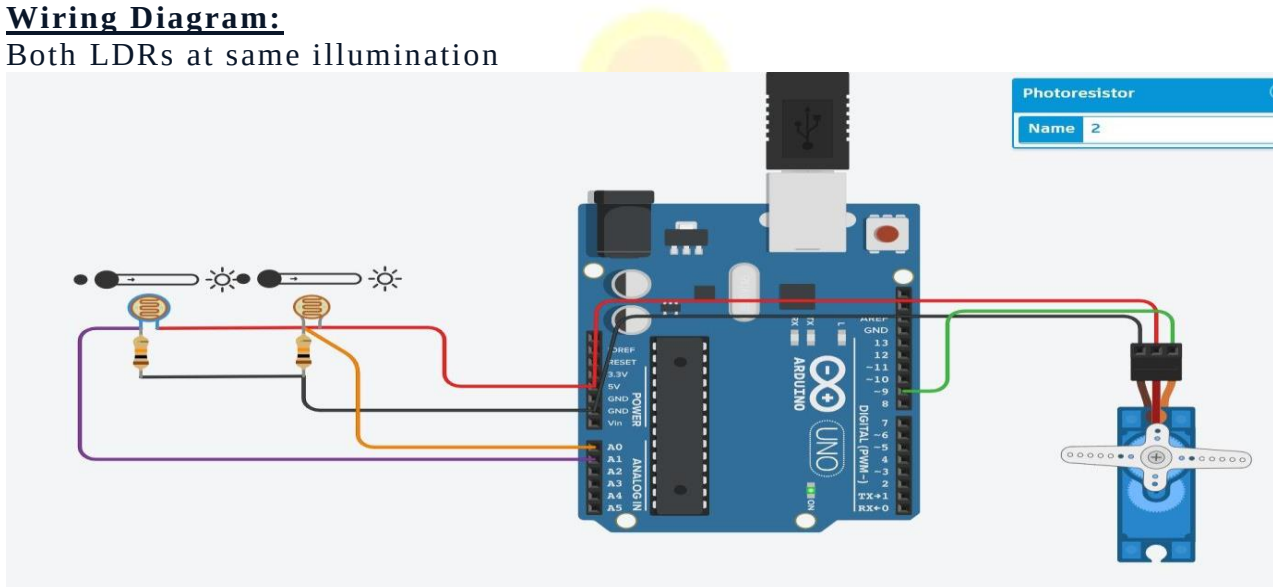
- $$\eta(\%) = \frac{P_{track} - P_{static}}{P_{static}} \times 100$$

3. Schematics:

Simulation on TinkerCAD

Wiring Diagram:

Both LDRs at same illumination



Component List

- **Microcontroller:** Arduino Uno R3
- **Actuator:** SG90 Micro Servo (responsible for the physical rotation of the solar panel).
- **Sensors:** Two Photoresistors (LDRs) used to detect differential light levels.
- **Passive Components:** Two **10kΩ Resistors** (acting as pull-down resistors for the voltage divider circuits).

Wiring Description

A. The Sensor Circuit (Voltage Dividers)

The sensors are wired in a dual **Voltage Divider** configuration to convert light resistance into a measurable voltage.

- **Power:** A red wire carries 5V from the Arduino to one leg of both LDRs.

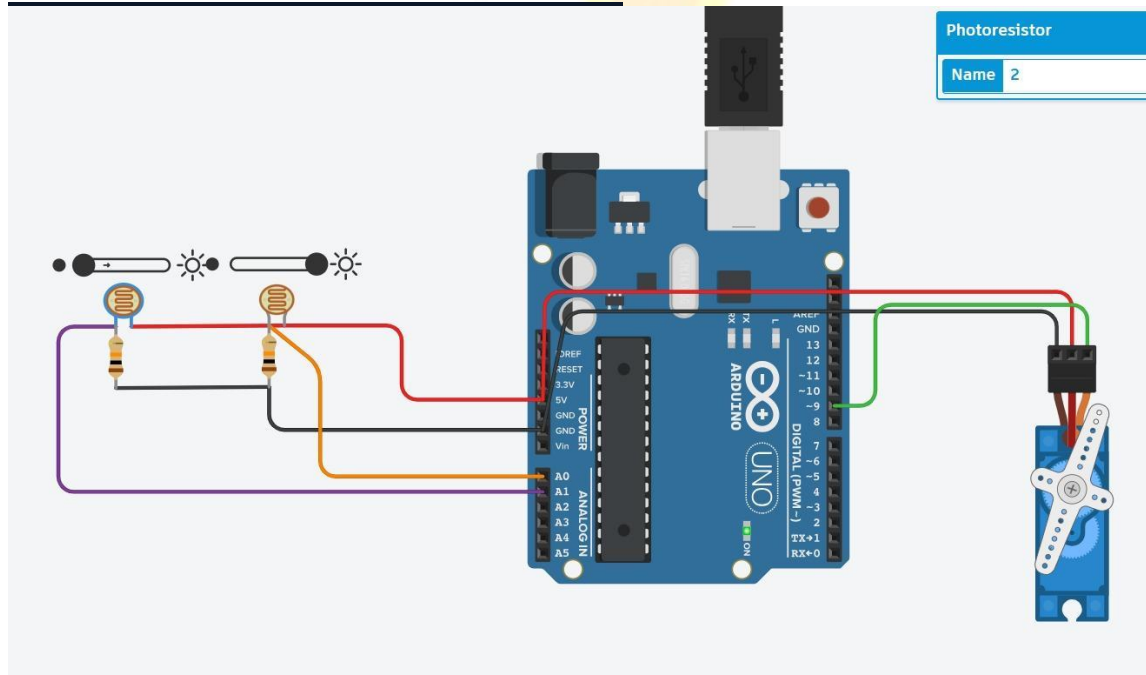
- **Signal:** The junction between each LDR and its corresponding $10k\Omega$ resistor is tapped for the signal.
 - The **Orange wire** connects the left sensor to Analog Pin **A0**.
 - The **Purple wire** connects the right sensor to Analog Pin **A1**.
- **Ground:** The other leg of both $10k\Omega$ resistors is connected to a common black wire leading to the Arduino's **GND** pin.

B. The Actuator Circuit (Servo Motor)

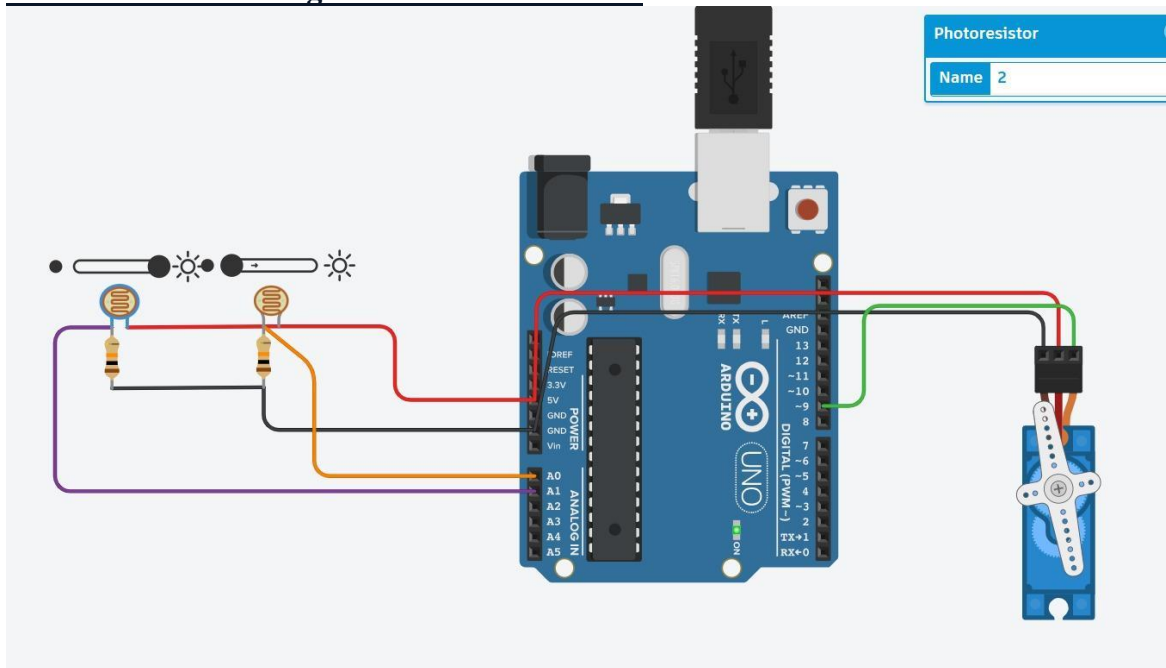
The servo motor is connected using a standard 3-wire interface:

- **Power (Red):** Connected to the **5V** rail to provide motor torque.
- **Ground (Black):** Connected to the **GND** pin to complete the circuit.
- **Signal (Green):** Connected to Digital **Pin 9**. This pin uses **Pulse Width Modulation (PWM)** to tell the servo exactly what angle (0° to 180°) to hold.

Simulation ON: Left LDR illuminated



Simulation ON: Right LDR illuminated



The simulation shows how a solar tracker automatically follows a light source using two light sensors (LDRs) and a servomotor.

- **How it moves:** The Arduino compares the light hitting the left and right sensors. If the **Right LDR** is brighter, the motor rotates **clockwise** (toward 0°). If the **Left LDR** is brighter, the motor rotates **anticlockwise** (toward 180°).
- **Staying Still:** To keep the motor from shaking or moving constantly, we included a "tolerance" in the code. This means if the light levels on both sides are almost the same, the motor stays still because it is already facing the light.

evening
position

noon
position

morning
position

E

Arduino Code:

```

1  #include <Servo.h>
2
3  Servo trackerServo;
4  int pos = 90;           // Start position (center 0-180)
5  int tolerance = 20;     // Minimum raw difference to trigger movement
6  int sensorWait = 15;
7
8  // Pin Definitions
9  const int ldrLeftPin = A0;
10 const int ldrRightPin = A1;
11 const int servoPin = 9;
12
13 void setup() {
14     Serial.begin(9600);
15     trackerServo.attach(servoPin);
16     trackerServo.write(pos);
17
18     Serial.println("Solar Tracker Initialized...");
19     delay(1000);
20 }
21
22 void loop() {
23     // 1. Read raw sensor values
24     int leftRaw = analogRead(ldrLeftPin);
25     int rightRaw = analogRead(ldrRightPin);
26
27     // 2. Convert raw values to Voltages (0.0V - 5.0V)
28     float leftVoltage = leftRaw * (5.0 / 1023.0);
29     float rightVoltage = rightRaw * (5.0 / 1023.0);
30
31     // 3. Calculate Error Voltage for your plot
32     float errorVolt = leftVoltage - rightVoltage;
33
34     // 4. Print formatted data (0-180 angle)
35     Serial.print("L_Volt: "); Serial.print(leftVoltage); Serial.print("V");
36     Serial.print(" | R_Volt: "); Serial.print(rightVoltage); Serial.print("V");
37     Serial.print(" | Error: "); Serial.print(errorVolt); Serial.print("V");
38     Serial.print(" | Angle: "); Serial.print(pos); Serial.println(" deg");
39
40     // 5. Movement Logic
41     int diff = leftRaw - rightRaw;
42
43     if (abs(diff) > tolerance) {
44         if (leftRaw > rightRaw && pos < 180) {
45             pos++;
46         }
47         else if (rightRaw > leftRaw && pos > 0) {
48             pos--;
49         }
50         trackerServo.write(pos);
51     }
52
53     delay(sensorWait);
54 }

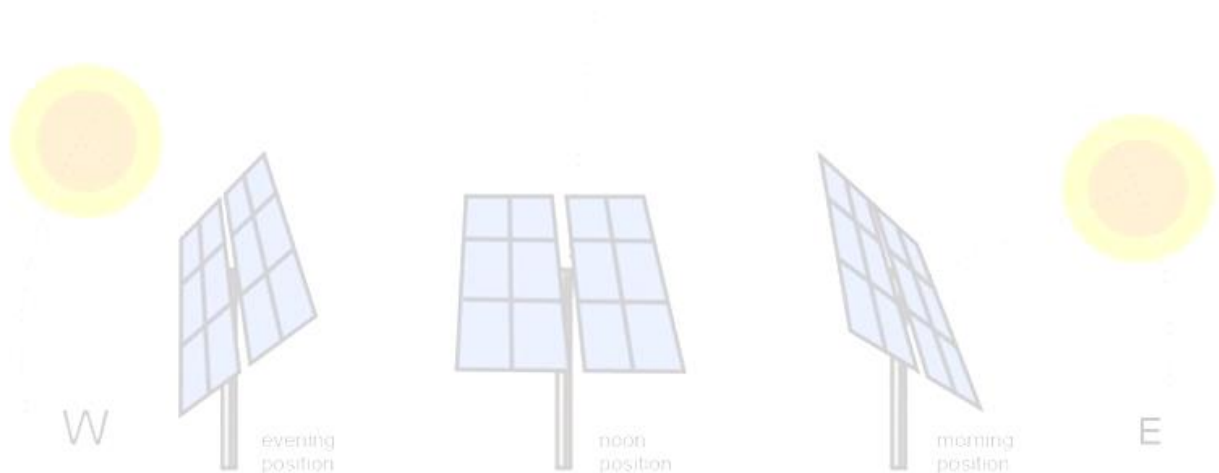
```



E

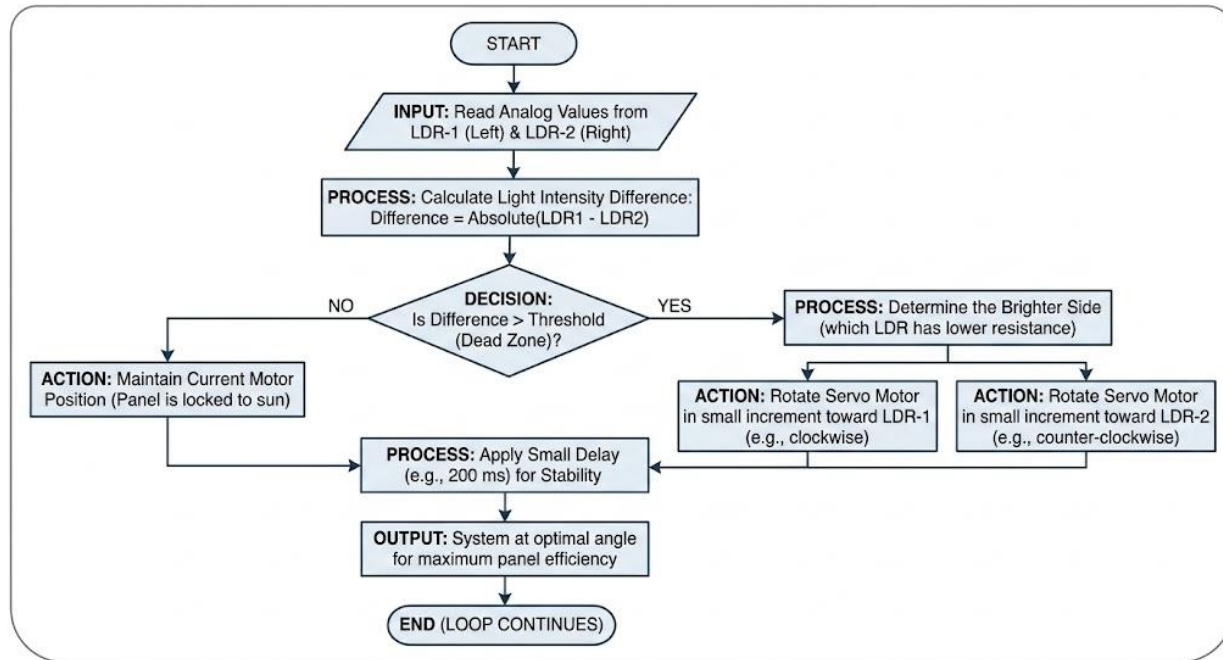
The Arduino code acts as the "brain" of the tracker, following a simple loop to keep the panel aligned with the light. It is divided into three sections:

- **Step 1: Sensor Input (Reading)** The code reads the voltage from the two LDRs on pins **A0** and **A1**. It then converts these raw numbers into a **Voltage (0V - 5V)** scale so we can monitor exactly how much light each sensor is receiving.
- **Step 2: Error Calculation (Deciding)** The program calculates the **difference** between the left and right light levels. It checks this difference against a **"Tolerance"** value. If the difference is small, the code does nothing (this prevents the motor from jittering). If the difference is large, it decides which side is brighter.
- **Step 3: Motor Output (Acting)** Based on which side is brighter, the code increases or decreases the **pos (position)** variable by 1 degree. It then sends this new angle to the **Servomotor** using the **trackerServo.write()** command. This process repeats every 15 milliseconds, creating smooth movement.



4. Working:

The solar tracking system operates as a closed-loop control system. The main objective is to keep the solar panel perpendicular to the sun's rays to ensure maximum energy absorption. The logical flow of the entire process is illustrated in the flowchart below:
Figure 1: Flowchart of the Solar Tracking Logic.



As

shown in the flowchart, the system continuously monitors the light intensity through sensors and adjusts the motor position accordingly. The detailed working steps are as follows:

1. Sensing Phase (Light Detection)

The system uses two **Light Dependent Resistors (LDRs)** placed on either side of the solar panel, separated by a divider (wedge). These sensors continuously monitor the intensity of incident sunlight. Since LDR resistance decreases as light intensity increases, the voltage levels at the analog pins of the microcontroller change accordingly.

2. Processing Phase (Comparison Logic)

The analog signals from the LDRs are sent to the **Microcontroller (e.g., Arduino)**. The controller executes a pre-defined algorithm to compare the values:

- **Case A (Left > Right):** If the left LDR receives more light, the controller detects a positive error.
- **Case B (Right > Left):** If the right LDR receives more light, the controller detects a negative error.
- **Case C (Left ≈ Right):** If the difference is within a small range (Threshold), the system remains stable to avoid unnecessary oscillations.

3. Actuation Phase (Mechanical Alignment)

Based on the comparison, the microcontroller sends a **Pulse Width Modulation (PWM)** signal to the **Servo Motor**. The motor rotates the solar panel in small increments (typically 1–2 degrees) towards the direction of maximum light. This ensures that the solar panel is always perpendicular to the sun's rays (90° angle of incidence), which is the point of maximum efficiency.

4. Continuous Tracking Loop

This is a real-time closed-loop system. As the sun moves across the sky from East to West, the sensors detect the shift in light, and the motor adjusts the panel position accordingly throughout the day. At night or in low-light conditions, the system enters a standby mode or returns to the initial position (depending on the code logic).

5.HARDWARE PICTURE:

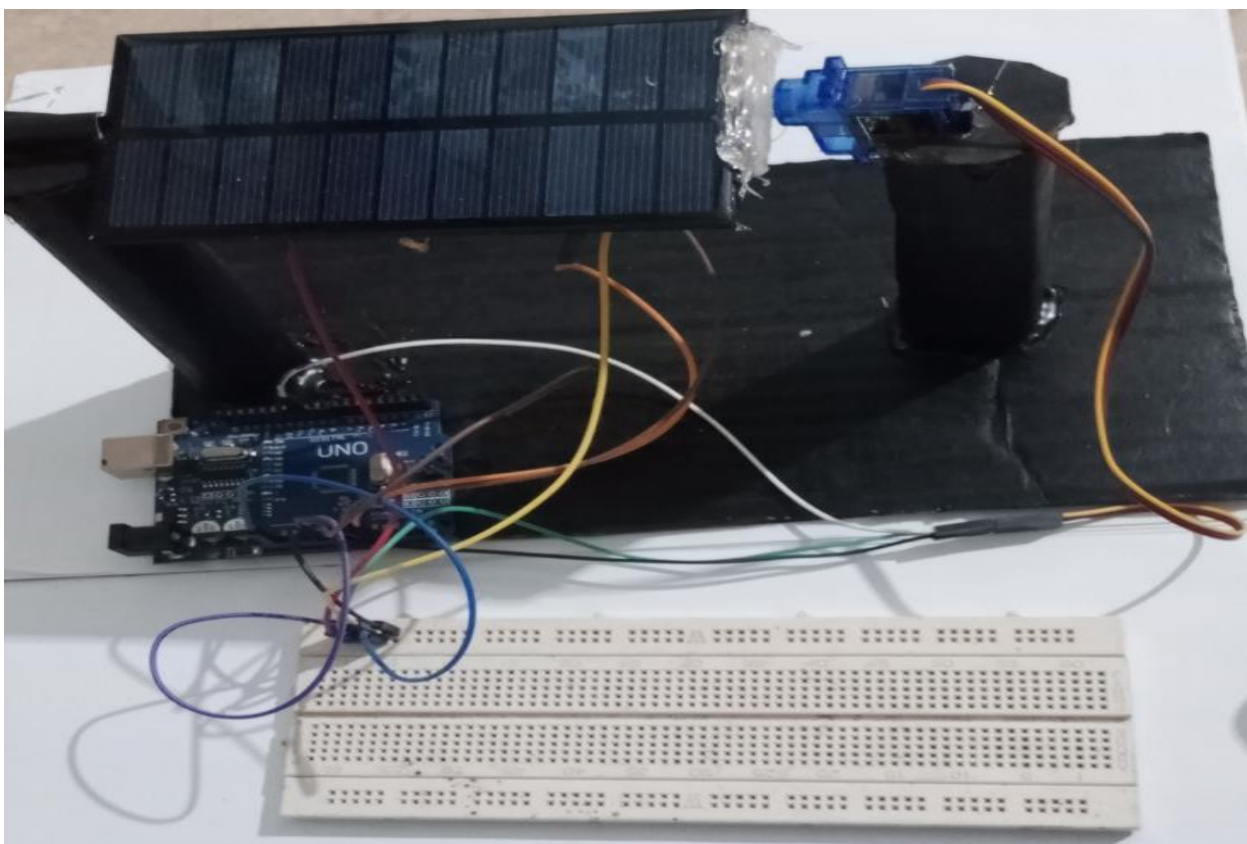


Figure 2: Hardware implementation of the project.

6.Key Operational Features:

The following technical aspects ensure the reliability and efficiency of the tracking system:

- **Precision Control (Threshold):** The system is programmed with a specific "Dead-zone" or threshold value. This prevents the motor from constantly jittering or oscillating due to very minor changes in light or environmental noise, thereby increasing the lifespan of the motor.

- **High Sensitivity LDRs:** We have used high-sensitivity Light Dependent Resistors that can detect even small shifts in the sun's position, ensuring that the panel remains perpendicular to the light source at all times.
- **Pulse Width Modulation (PWM):** The microcontroller uses PWM signals to control the servo motor, which allows for smooth and precise angular movements (typically 1–2 degrees per step).
- **Power Optimization:** The energy consumed by the tracking mechanism is significantly lower than the additional energy gained (approximately 25% to 40% increase) by keeping the panel aligned with the sun.

5. OBSERVATION

The following observations were recorded by systematically varying the incident light angle and intensity on the dual-LDR sensing module. All electrical parameters were measured using a digital multimeter. Graphs are to be plotted from the tabulated data below.

4.1 Measured Parameters — Data Table

#	Parameter	Symbol	Min	Max	Unit	Remarks
01	LDR Resistance – Full Sunlight	R_light	1	10	kΩ	Photoconductive effect
02	LDR Resistance – Dark/Shadow	R_dark	0.5	10	MΩ	High resistance state
03	Voltage Divider Output (Left LDR)	V_A0	0	5	V	Analog pin A0
04	Voltage Divider Output (Right LDR)	V_A1	0	5	V	Analog pin A1
05	Differential Error Voltage	ΔV	-5	+5	V	$V_{A0} - V_{A1}$
06	Servo Motor Angle	θ	0	180	°	PWM-controlled

Table 4.1 — Observed electrical parameters of the single-axis solar tracking system at various operating conditions

4.2 Working Formulas

All calculated values in the observation are derived using the following formulas:

Quantity	Formula
LDR Voltage Divider	$V_{out} = V_{cc} \times [R_{fixed} / (R_{LDR} + R_{fixed})]$
LDR Resistance Model	$R(E) = R_0 \cdot E^{(-\gamma)}$
Error / Tracking Signal	$\Delta V = V_{A0} - V_{A1}$

Quantity	Formula
Servo Angle Mapping	$\theta = \text{map}(\Delta V, -5 \text{ V}, +5 \text{ V}, 0^\circ, 180^\circ)$

4.4 Characteristic Curve — Differential Voltage vs. Panel Angle

Rotate the panel from 0° to 180° and record the differential error voltage $\Delta V = V_{A0} - V_{A1}$. The zero-crossing identifies the optimal sun-facing angle. This validates the null-balance tracking logic of the Arduino.

Illuminating LDRs:

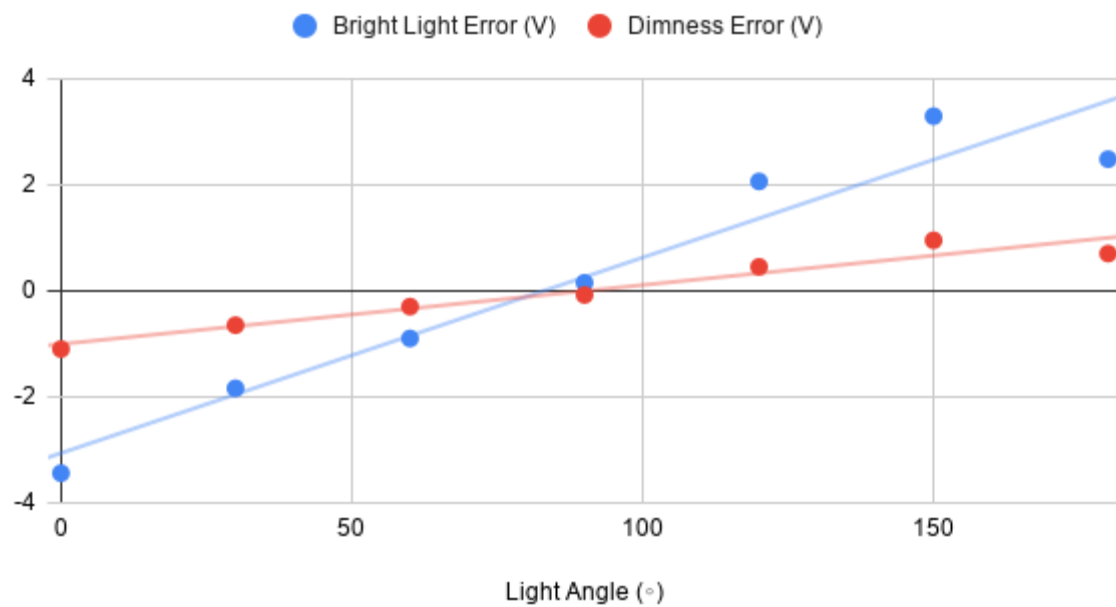
Light Angle ($^\circ$)	VLeft (V)	VRight (V)	Error Voltage (VDiff)
0	1.21	4.65	-3.44
30	1.5	3.34	-1.84
60	3.59	4.49	-0.9
90	4.86	4.6	0.26
120	4.83	3.59	1.24
150	4.52	1.23	3.29
180	3.69	1.21	2.48

Dimming LDRs:

Light Angle ($^\circ$)	VLeft (V)	VRight (V)	Error Voltage (VDiff)
0	0.85	1.95	-1.1
30	1.2	1.85	-0.65
60	2.1	2.4	-0.3
90	2.65	2.71	-0.06
120	2.8	2.35	0.45
150	2.9	1.95	0.95
180	2.6	1.9	0.7

Error/Differential Voltage Table:

Light Angle (°)	Bright Light Error (V)	Dimness Error (V)
0	-3.44	-1.1
30	-1.84	-0.65
60	-0.9	-0.3
90	0.26	-0.06
120	1.24	0.45
150	3.29	0.95
180	2.48	0.7

Bright Light Error (V) and Dim Light Error (V)*Figure — Differential LDR voltage vs. panel orientation. Zero-crossing = optimal tracking position*

6.Results:

Significance of the Zero-Crossing

The **Zero-Crossing** (the point where Error approximately = 0V) represents the **System Setpoint**.

- **Observed Data:** The system achieved a null balance between -0.06V and +0.26V at a physical angle of 90 degrees.
- **Significance:** This confirms that the differential sensor array is correctly aligned. The slight offset from a perfect 0.00V is attributed to hardware tolerances (resistor variance and LDR sensitivity mismatch). In the control loop, this crossing point tells the Arduino that the tracker is perfectly aligned with the light source and should cease movement.

Intensity vs. Sensitivity (The Two Lines)

The comparison between the **Bright** and **Dim** light tables reveals the **System Gain**.

- **Observation:** The "Bright" curve exhibits a significantly steeper slope compared to the "Dim" curve.
- **Significance:** This demonstrates that the tracker's sensitivity is proportional to ambient light intensity. In bright conditions, a 5 degree shift in light creates a large voltage error, leading to a fast corrective response. in dim conditions, the "Error Signal" is weaker, which explains the "sluggish" behavior sometimes observed in low-light environments.

The experimental data successfully validates the solar tracking algorithm through a **linear error characteristic**. This linear relationship confirms that the differential LDR array provides a **constant system gain**, ensuring a predictable and stable response across the full tracking range.

